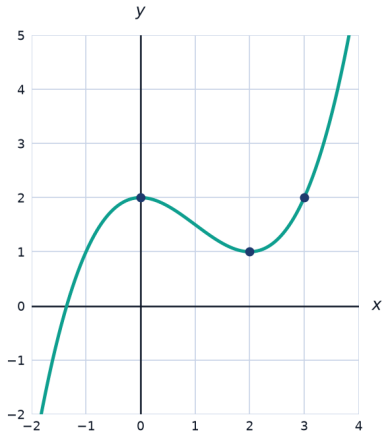
Rational Expressions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Simplifying radical expressions

For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(5)$.

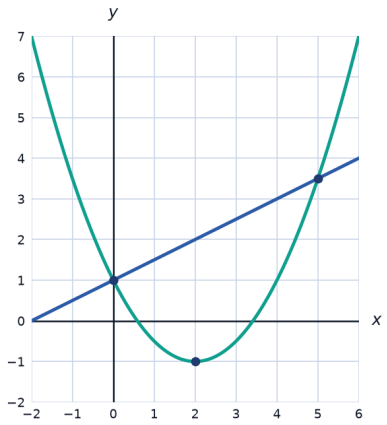


2 [Foundational] • Target: Simplifying radical expressions

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

3 [Foundational] • Target: Operations with radical expressions

For the model $f(x) = 4x + 7$ in operations with radical expressions, evaluate $f(2)$.



4 [Foundational] • Target: Operations with radical expressions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

5 [Foundational] • Target: Operations with radical expressions

A data set recorded for operations with radical expressions consists of the four values $\{15, 19, 23, 29\}$.

What is the arithmetic mean? Show your complete algebraic work.

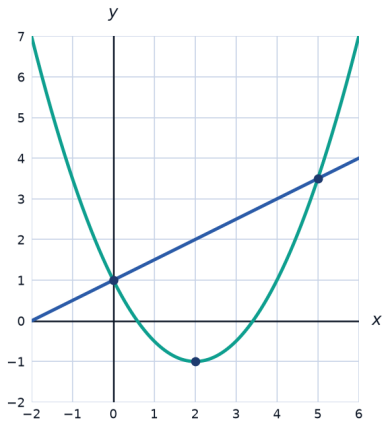
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6 [Foundational] • Target: Operations with radical expressions

For the model $f(x) = 4x + 7$ in operations with radical expressions, evaluate $f(3)$.

7 [Foundational] • Target: Operations with radical expressions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Operations with radical expressions

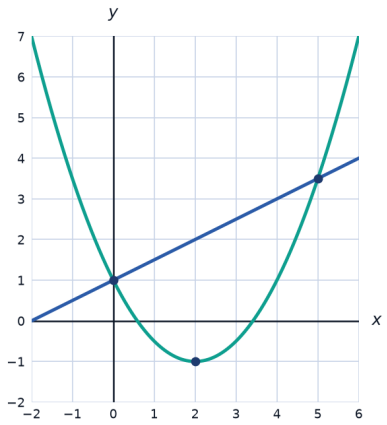
For the model $f(x) = 4x + 7$ in operations with radical expressions, evaluate $f(4)$.

9 [Intermediate] • Target: Operations with radical expressions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

10 [Intermediate] • Target: Operations with radical expressions

For the model $f(x) = 4x + 7$ in operations with radical expressions, evaluate $f(5)$.



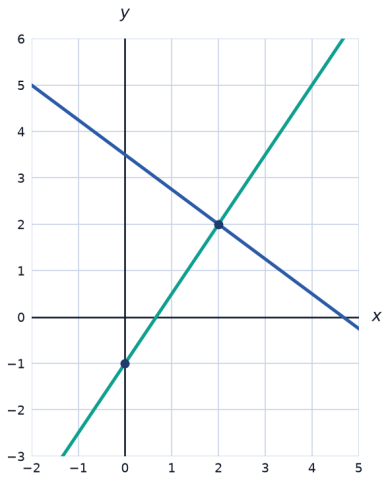
11 [Intermediate] • Target: Operations with radical expressions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

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12 [Intermediate] • Target: Graphs and characteristics of square root functions

For the model $f(x) = 5x + 8$ in graphs and characteristics of square root functions, evaluate $f(2)$.



13 [Intermediate] • Target: Graphs and characteristics of square root functions

A data set recorded for graphs and characteristics of square root functions consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

14 [Intermediate] • Target: Graphs and characteristics of square root functions

For the model $f(x) = 5x + 8$ in graphs and characteristics of square root functions, evaluate $f(3)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

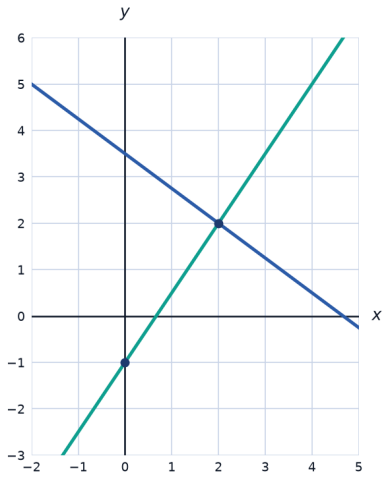
15 [Challenging] • Target: Graphs and characteristics of square root functions • Cross-Lesson: 9.01 Trigonometric Functions

For the model $f(x) = 5x + 8$ in graphs and characteristics of square root functions, evaluate $f(4)$.

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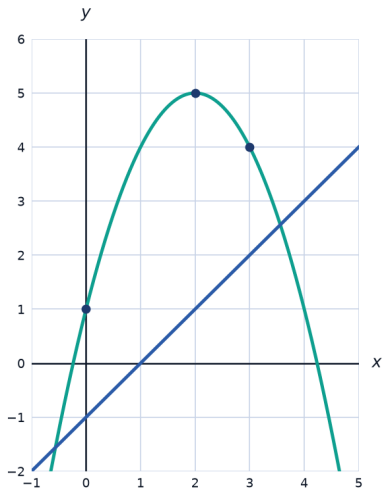
16 [Challenging] • Target: Graphs and characteristics of square root functions • Cross-Lesson: 1.01 Functions & Their Graphs

For the model $f(x) = 5x + 8$ in graphs and characteristics of square root functions, evaluate $f(5)$.



17 [Challenging] • Target: Graphs and characteristics of cube root functions • Cross-Lesson: 10.01 Matrices & Systems

For the model $f(x) = 6x + 9$ in graphs and characteristics of cube root functions, evaluate $f(2)$.



18 [Challenging] • Target: Graphs and characteristics of cube root functions • Cross-Lesson: 11.01 Probability & Statistics

A data set recorded for graphs and characteristics of cube root functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

19 [Challenging] • Target: Graphs and characteristics of cube root functions • Cross-Lesson: 2.01 Quadratics & Complex Numbers

For the model $f(x) = 6x + 9$ in graphs and characteristics of cube root functions, evaluate $f(3)$.

20 [Challenging] • Target: Graphs and characteristics of cube root functions • Cross-Lesson: 3.01 Polynomial Functions

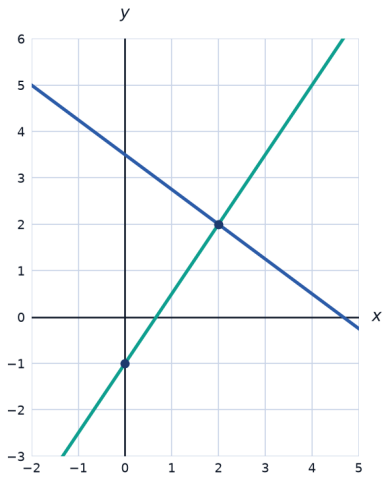
For the model $f(x) = 6x + 9$ in graphs and characteristics of cube root functions, evaluate $f(4)$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Review: Laws of exponents

For the model $f(x) = 6x + 4$ in review: laws of exponents, evaluate $f(2)$.



a. 16

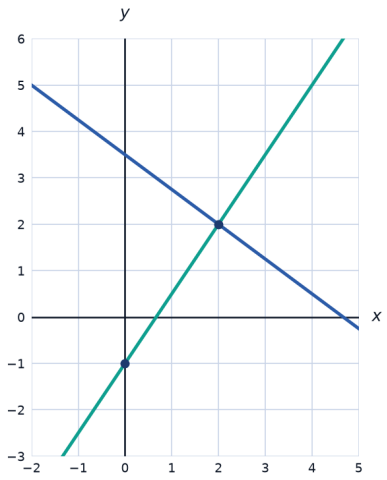
b. 22

c. 12

d. 20

22 [Foundational] • Target: Review: Laws of exponents

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

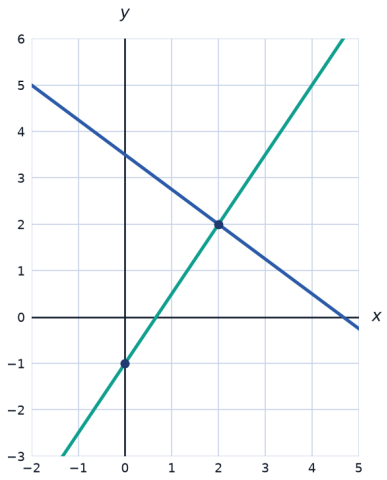
c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

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23 [Foundational] • Target: Review: Laws of exponents

For the model $f(x) = 6x + 4$ in review: laws of exponents, evaluate $f(5)$.



a. 34

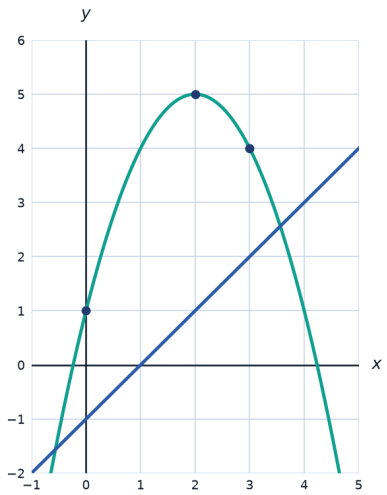
b. 40

c. 30

d. 38

24 [Foundational] • Target: Fractional powers

For the model $f(x) = 2x + 5$ in fractional powers, evaluate $f(2)$.



a. 9

b. 11

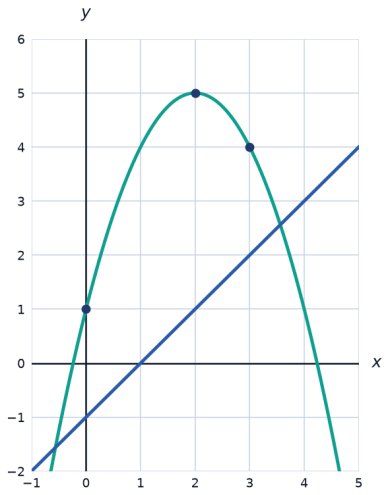
c. 4

d. 13

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25 [Foundational] • Target: Fractional powers

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

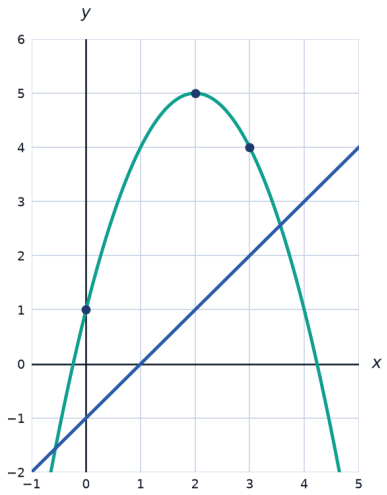
b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

26 [Foundational] • Target: Fractional powers

For the model $f(x) = 2x + 5$ in fractional powers, evaluate $f(5)$.



a. 15

b. 17

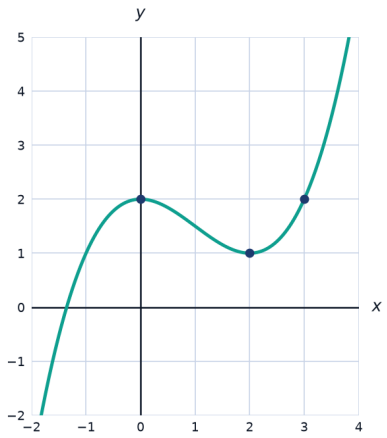
c. 10

d. 19

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27 [Foundational] • Target: Simplifying radical expressions

For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(2)$.



a. 12

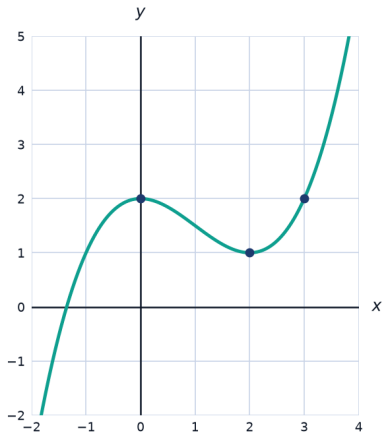
b. 15

c. 6

d. 16

28 [Foundational] • Target: Simplifying radical expressions

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Review: Laws of exponents

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

30 [Intermediate] • Target: Review: Laws of exponents

A data set recorded for review: laws of exponents consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

31 [Intermediate] • Target: Review: Laws of exponents

For the model $f(x) = 6x + 4$ in review: laws of exponents, evaluate $f(3)$.

a. 22

b. 28

c. 18

d. 26

32 [Intermediate] • Target: Review: Laws of exponents

For the model $f(x) = 6x + 4$ in review: laws of exponents, evaluate $f(4)$.

a. 28

b. 34

c. 24

d. 32

33 [Intermediate] • Target: Review: Laws of exponents

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

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34 [Intermediate] • Target: Review: Laws of exponents

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

35 [Intermediate] • Target: Fractional powers

A data set recorded for fractional powers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

36 [Intermediate] • Target: Fractional powers

For the model $f(x) = 2x + 5$ in fractional powers, evaluate $f(3)$.

a. 11

b. 13

c. 6

d. 15

37 [Intermediate] • Target: Fractional powers

For the model $f(x) = 2x + 5$ in fractional powers, evaluate $f(4)$.

a. 13

b. 15

c. 8

d. 17

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Fractional powers • Cross-Lesson: 11.01 Probability & Statistics

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

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39 [Challenging] • Target: Fractional powers • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

40 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 3.01 Polynomial Functions

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

41 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(3)$.

a. 15

b. 18

c. 9

d. 19

42 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 6.01 Radicals & Rational Exponents

For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(4)$.

a. 18

b. 21

c. 12

d. 22

43 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 7.01 Sequences & Series

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

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44 [Challenging] • Target: Operations with radical expressions • Cross-Lesson: 8.01 Conic Sections

A data set recorded for operations with radical expressions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

45 [Challenging] • Target: Graphs and characteristics of square root functions • Cross-Lesson: 9.01 Trigonometric Functions

A data set recorded for graphs and characteristics of square root functions consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

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